

Ashutosh Jha

PhD Student, Machine Learning
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Email: a.jha@tu-braunschweig.de

Braunschweig, Germany

Languages: English (C1), Deutsch (B1)

Education

PhD in Machine Learning

Sep 2026 - Present

Institute of Artificial Intelligence, TU Braunschweig, Germany

Advisor: Prof. Dr. Michel Besserve

Causal and probabilistic representation learning for complex systems.

M.Sc. Quantitative Data Science Methods, Machine Learning & Econometrics

Oct 2023 - Aug 2026

Eberhard Karls Universität Tübingen, Germany

Grade: 1.27/5.0 (1.0 = best)

Thesis: Optimal Transport in linear Independent Component Analysis, at MPI-IS Tübingen. Thesis grade: 1.0

Advisors: Dr. Simon Buchholz, Prof. Dr. Michel Besserve, Prof. Dr. Joachim Grammig

Erasmus Exchange Semester: Econometrics and Machine Learning

Feb 2025 - Jun 2025

Katholieke Universiteit Leuven, Belgium

Grade: 1.5/5.0 (1.0 = best)

B.E. Electrical and Electronics Engineering

Aug 2016 - May 2020

Birla Institute of Technology and Science, Pilani, Goa Campus, India

Grade: 8.12/10

Publications

Linear Independent Component Analysis via Optimal Transport

Ashutosh Jha, Michel Besserve, Simon Buchholz.

9th Workshop on Tractable Probabilistic Modeling (TPM), UAI 2026, Amsterdam.

[arXiv:2607.14081](#) | [OpenReview](#) | [PDF](#)

- We replace ICA's proxy contrast functions with the squared L^2 Wasserstein distance to a standard Gaussian, computable by sorting, prove it is maximized at independent components, and show it outperforms classical methods on heterogeneous source distributions.

Research Experience

Student Research Assistant, Perceiving Systems

Nov 2025 - Jun 2026; Jul 2024 - Jun 2025

at [Max Planck Institute for Intelligent Systems](#), Tübingen, Germany

- Extended the NICE toolbox to process non-verbal communication data by integrating computer-vision detectors for pose estimation, emotion detection, and head orientation. Tools: Python, PyTorch.
- Containerised the application and managed CI/CD pipelines to make experimental models reproducible across machines. Tools: Docker, Apptainer, GitLab CI/CD, Hugging Face, Linux.

Quantitative Analyst Intern, Financial Stability

Jul 2025 - Sep 2025

at [Deutsche Bundesbank](#), Frankfurt, Germany

- Modelled investment-fund portfolio vulnerabilities against market, monetary-policy, and geopolitical shocks by stress-testing econometric, probabilistic decision-tree, and foundation tabular (TabPFN) time-series models. Tools: Python, SQL, Git.

Industry Experience

Software Engineer, HSBC Holdings PLC, Pune, India

Sep 2020 - Sep 2023

Cloud economics and FinOps: statistical cost-benefit models (\$500K savings, 1.5× vendor discounts) and real-time ETL pipelines at 4× lower latency. Python, SQL, GCP, AWS, Jenkins.

Software Engineering Intern, Nvidia Corporation, Bengaluru, India

Aug 2019 - Dec 2019

Real-time visualisation tool for root-cause analysis of SoC CPU boot-sequence verification. Python, Linux, shell scripting.

Selected Projects

Structured Product Design, Pricing and Hedging, KU Leuven *Feb 2025 - May 2025*

Git Repository: priced and hedged a Bonus Certificate under the Bates model (stochastic volatility with jumps), using a two-stage calibration and Monte Carlo simulation for the path-dependent payoff.

Probabilistic Asset Pricing, Master Seminar, Universität Tübingen *Nov 2024 - Jan 2025*

Git Repository: Bayesian asset pricing using asset-pricing theory as a prior and a state-space representation (Kalman filters) to estimate posterior prices, anomalies, and risk premia.

Awards

Best Pitch & Co-creation Prize, Neckarthon, Startup Center Tübingen *Nov 2024*

Dual prize for “Tue-bi-smart”, an ML-enabled transport prototype for Stadtwerke Tübingen.

Prime Minister’s Scholarship Scheme, Government of India *2017*

Awarded for academic excellence during undergraduate studies.

Technical Skills

Machine Learning & Research: *PyTorch, deep learning, probabilistic & Bayesian modelling, optimal transport, causal inference, time-series forecasting, LLM agents (LangChain, LangGraph), Hugging Face, TabPFN.*

Programming & Databases: *Python, SQL, R, MATLAB, VBA.*

MLOps & Tools: *Git, Docker, Apptainer, CI/CD (GitLab CI/CD, Jenkins), HTCondor, Linux, LaTeX.*

Cloud & BI: *GCP (Compute, BigQuery, Cloud Run), AWS (Lambda, S3), Power BI, MS Excel.*

Relevant Coursework: *Machine Learning, Deep Learning, Probabilistic Machine Learning, Optimization, Bayesian Modelling, Advanced Time Series Analysis, Machine Learning in Econometrics, Financial Economics.*